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Evaluating large language models as grant reviewers: a comparative study of prompt engineering strategies

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Grant application review is resource-intensive and subject to inter-rater variability. Large language models (LLMs) may augment this process, but their reliability in grant evaluation remains unexplored. This exploratory pilot study compared LLM-generated grant reviews to human expert reviews across three prompt engineering strategies: zero-shot (no examples), few-shot (multiple training examples), and few-shot with strict scoring instructions. Three commercial LLMs (OpenAI GPT-5 Nano, Google Gemini 2.5 Flash, xAI Grok 4 Fast Reasoning) each reviewed three de-identified faculty seed grant applications (\$5,000 awards) under each condition, generating 63 LLM reviews compared against 12 human expert reviews. Among the three conditions tested, few-shot prompting produced scores closest to human reviewers (mean difference: -2.57 points on a 100-point scale, $p = 0.671$, Glass's $\Delta = -0.19$). Zero-shot prompting showed moderate optimism bias (+4.55 points, $p = 0.347$) and strict instructions produced substantial under-scoring (-10.68 points, $p = 0.097$). LLMs demonstrated 49%–59% lower score dispersion than humans (49% in Experiment 1 and 59% in Experiments 2–3) likely reflects range restriction. Chance-corrected agreement between LLM and human funding recommendations was at or below chance in every condition (Cohen's κ : -0.11–0.05), indicating LLM funding recommendations provided no meaningful signal beyond chance. These preliminary findings suggest prompt engineering affects LLM scoring behavior in this sample, and that LLMs cannot yet reliably replicate human grant review scores or funding recommendations but may provide adjunctive data for consideration.

KEYWORDS

artificial intelligence, grant review, large language models, peer reviewer, prompt engineering

1 Introduction

The evaluation of research grant applications is a cornerstone of academic funding, yet it remains resource-intensive, placing substantial demands on expert reviewers (Guthrie et al., 2019; National Institutes of Health, 2024). Studies have consistently identified bias and inter-rater inconsistency as persistent limitations of peer review (Marsh et al., 2008; Pier et al., 2018). Large language models (LLMs) have demonstrated consistency in applying structured evaluation criteria (Zhang et al., 2024) and may reduce the burden of initial screening in peer review workflows (Checco et al., 2021), suggesting potential advantages for grant review. However, significant concerns remain about whether LLMs can capture the nuanced judgment that experienced human reviewers bring to holistic evaluation, and prior research has documented “optimism bias” in LLM evaluations (Liang et al., 2024).

TABLE 1 Summary of experimental design.

Application pool	Experiment 1: Zero-shot	Experiment 2: Few-shot	Experiment 3: Few-shot + strict
FC1	Test data	Test data	Test data
FC2	Test data	Test data	Test data
FC3	Test data	Training only ^a	Training only ^a
Total LLM reviews	27 reviews	18 reviews	18 reviews
Human comparison	12 reviews	8 reviews	8 reviews

^aApplications labeled as Test Data were reviewed and scored by the models. The application labeled Training Only (FC3) was used solely to provide the models with examples of human reviews showing natural score variance; it was not scored by the models in those experiments.

A growing body of evidence suggests that LLM performance is highly sensitive to prompt engineering: the design of instructions and examples provided to guide model behavior (Sahoo et al., 2024; Wei et al., 2022). Despite the recognized importance of prompt design, no prior study has systematically compared prompt engineering strategies in the context of grant review. While several studies have explored LLM applications in journal manuscript peer review (Hosseini and Horbach, 2023; Thelwall, 2024), the grant review context presents unique challenges: assessment across diverse criteria (innovation, methodology, team, budget, feasibility), evaluation of proposed rather than completed research, and decisions with direct financial implications.

This exploratory pilot study investigated whether LLMs can generate grant review scores that align with human expert reviewers and how different prompt engineering strategies affect this alignment. We tested three conditions using a grading rubric: (1) zero-shot prompting, (2) few-shot prompting using multiple human review examples demonstrating natural score variance, and (3) few-shot prompting combined with explicit strict scoring instructions. Three commercial LLMs [OpenAI GPT-5 Nano (OpenAI, 2025), Google Gemini 2.5 Flash (Google DeepMind, 2025), xAI Grok 4 Fast Reasoning (xAI, 2025)] were evaluated across all conditions using applications submitted for an institutional seed grant program.

2 Methods

2.1 Study design

We conducted a multi-condition, multi-vendor comparative evaluation at Stony Brook University School of Health Professions using applications from the Research Seed Grant Program (awards up to \$5,000). This study was approved by the Stony Brook University IRB (Protocol: IRB2025-00534, classified as non-human subjects research). Four applications were submitted in the funding cycle under study. All four applicants were invited to participate, with one applicant excluded due to declined consent, and the remaining three retained.

2.2 Grant applications and human reviews

Three de-identified applications (pseudonymized as FC1, FC2, and FC3) were each independently reviewed by four expert reviewers ($n = 12$ total human reviews). Reviewers held doctoral degrees with minimum 5 years of research experience and prior

grant review experience. All reviews used a standardized six-criterion rubric totaling 100 points: Innovation and Impact (30 points), Research Methods (30 points), Team Qualifications (10 points), Funding Potential (10 points), Budget (10 points), and Writing Quality (10 points). Reviewers also provided a funding recommendation: Fund, Fund with Revisions, or Do Not Fund. The full rubric is provided in [Supplementary Appendix A](#).

2.3 Experimental conditions

Three prompt engineering strategies were tested. For each experiment evaluation instructions, the scoring rubric and the seed grant application text were uploaded for use by the LLM. The complete prompt templates for all three conditions are provided in [Supplementary Appendix B](#) and in machine-readable form in the [Supplementary Material \(S1\)](#). To prevent data contamination in experiments using training examples (Experiments 2–3), FC3’s application and its four human reviews served as training data and were excluded from all comparisons in those experiments.

Experiment 1 (Zero-Shot): LLMs received no training examples. All 12 human reviews served as the comparison group (3 applicants $\times$ 4 reviews) for the LLM reviews (27 LLM reviews: 3 vendors $\times$ 3 applicants $\times$ 3 iterations).

Experiment 2 (Few-Shot): LLMs received all four human reviews of FC3’s application as training examples demonstrating natural score variance (range: 61–80 points). Only FC1 and FC2 reviews were compared with the LLM reviews (18 LLM reviews: 3 vendors $\times$ 2 applicants $\times$ 3 iterations vs. 8 human reviews: 2 applicants $\times$ 4 reviews).

Experiment 3 (Few-Shot + Strict): LLMs received the same training examples as Experiment 2, supplemented with explicit instructions to apply conservative scoring (e.g., “Apply high standards”, “Reserve high scores for truly exceptional proposals”, “Be critical in your assessment”).

Only FC1 and FC2 reviews were compared with the LLM reviews (18 LLM reviews: 3 vendors $\times$ 2 applicants $\times$ 3 iterations vs. 8 human reviews: 2 applicants $\times$ 4 reviews).

2.4 LLM configuration

Three commercial LLMs were evaluated using October 2025 API versions: OpenAI GPT-5 Nano (gpt-5-nano-2025-08-07),

Google Gemini 2.5 Flash (gemini-2.5-flash-preview-09; Google DeepMind, 2025), and xAI Grok 4 Fast Reasoning (grok-4-fast-reasoning; the non-reasoning variant was not used). xAI Grok used temperature = 0.1 and max tokens = 4,000; OpenAI and Google models used default API parameters (documented default temperature = 1.0 for both vendors at the time of data collection). This inconsistency in temperature settings across vendors is a limitation of the study design. Differences in temperature confound vendor-level comparisons and analyses, as vendor differences may reflect differences in temperature rather than differences due to the underlining model. Vendor-level results (Section 3.3) should therefore be interpreted as descriptive only, and low temperature setting for xAI Grok may contribute to lower output variability for that vendor. No system prompts were used beyond the experimental prompt templates. Three independent iterations per vendor-application-experiment combination yielded 63 LLM reviews across the three experiments (27 in Experiment 1; 18 each in Experiments 2–3 after excluding the training application). No LLM calls failed or were excluded; one criterion-level value (Team Strength) was missing from a single review but the total score was still produced.

2.5 Statistical analysis

Given the small sample size and the nested structure of the data, all statistical analyses should be interpreted as exploratory rather than confirmatory. Each LLM produced three non-independent repeated outputs per application, so these iterations were averaged within each vendor-application combination to yield a single representative score, resulting in 9 independent LLM observations in Experiment 1 (3 vendors × 3 applications), and 6 LLM observations in Experiments 2 and 3 (3 vendors × 2 applications). Within-experiment *t*-tests compared these averaged LLM scores against individual human scores. Effect sizes were calculated using Glass’s Δ, which uses the human group’s standard deviation as the baseline; this measure was chosen because LLM scores showed roughly 2.5 times less variability than human scores. We report 95% confidence intervals for mean differences; confidence intervals for Glass’s Δ were computed via bias-corrected bootstrap (10,000 resamples). To address non-independence of human scores (four reviewers each scored multiple applications), we fit a linear mixed-effects model for each experiment with a fixed effect of reviewer type (LLM vs. human) with random intercepts for rater identity (human reviewer or LLM vendor) and application.

Human scores are treated as independent observations in the *t*-tests, but because all four reviewers each scored all applications, correlation was introduced among scores from the same reviewer that a *t*-test does not account for. Thus a linear mixed-effects models described above was also used to quantify the effect of this violation. To compare LLM scores across prompt conditions, a one-way ANOVA was restricted to the two applications common to all experiments (FC1 and FC2) to ensure balanced group sizes. Criterion-level analyses are exploratory. To control for multiple comparisons across the six scoring criteria within each experiment, Holm-Bonferroni correction was applied, and both uncorrected and corrected *p*-values are reported.

Recommendation agreement was assessed using Cohen’s κ, a chance-corrected measure of agreement at the exact category level (Fund/Fund with Revisions/Do Not Fund) and the broad level (positive vs. negative funding direction). To avoid non-independent pairs, each vendor-application combination was first reduced to a single LLM recommendation defined as the most frequent recommendation across its three iterations (matching the aggregation used for scores). Each of these aggregated LLM recommendations was then paired with each individual human review of the same application, yielding 36 pairs in Experiment 1 and 24 pairs in Experiments 2–3. Finally, a sensitivity computation over all raw human-LLM review pairs is also reported for comparison.

All analyses were conducted in Python 3.13 using the following python packages: scipy, statsmodels, scikit-learn, and pandas.

3 Results

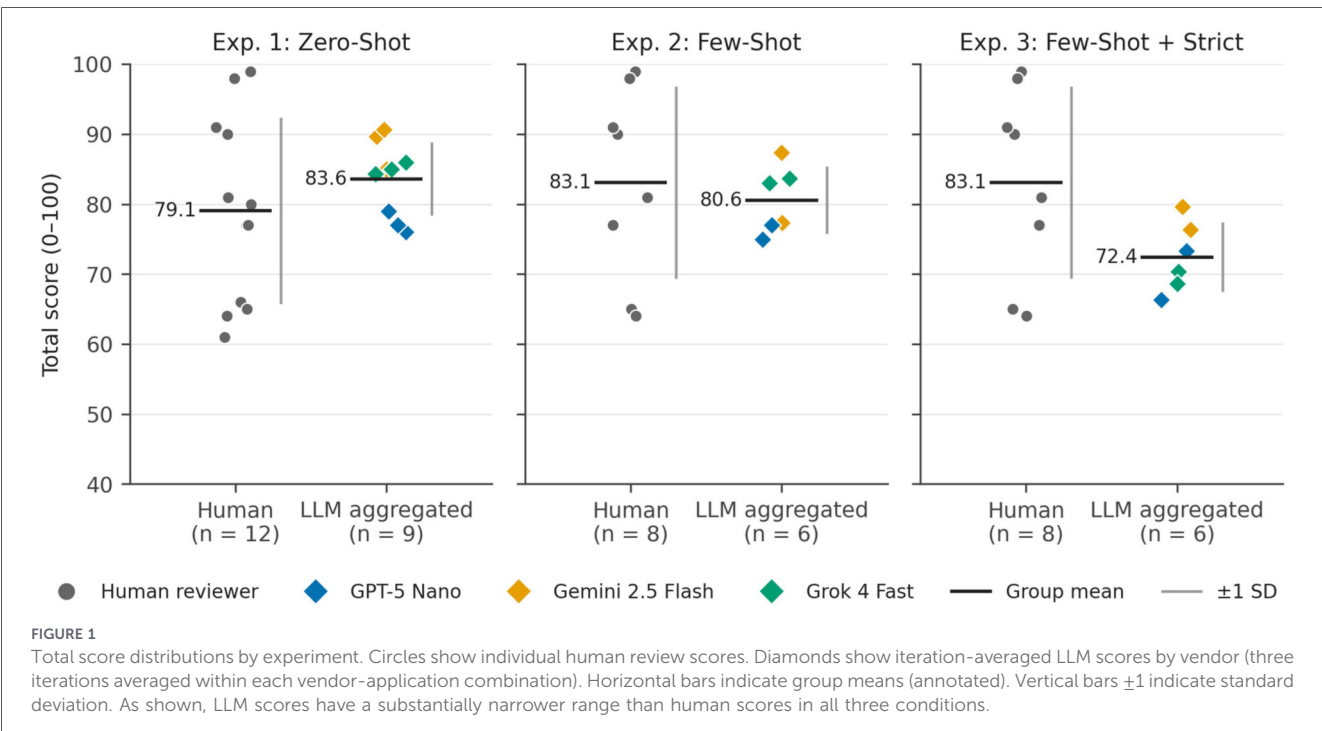
3.1 Overall score alignment

Table 2 summarizes results across all three experiments. Figure 1 displays the underlying score distributions. All LLM statistics use averaged scores (aggregated across 3 iterations per vendor-application combination) as described in Section 2.5. In the zero-shot condition (Experiment 1), LLMs scored applications 4.55 points higher than humans [83.63 ± 5.24 vs.

TABLE 2 Summary of experimental results across three prompt engineering conditions.

Metric	Exp1: Zero-shot	Exp2: Few- shot	Exp3: Few- shot & strict
Human reviews (<i>n</i>)	12	8	8
LLM reviews (<i>n</i> raw/aggregated)	27/9	18/6	18/6
Human mean ± SD	79.08 ± 13.32	83.13 ± 13.73	83.13 ± 13.73
LLM mean ± SD (aggregated)	83.63 ± 5.24	80.56 ± 4.81	72.44 ± 4.98
Mean difference [95% CI]	+4.55 [−3.36, +12.33]	−2.57 [−11.83, +6.97]	−10.68 [−20.11, −1.03]
<i>p</i> -value (aggregated)	0.347	0.671	0.097
Glass’s Δ [95% CI]	0.34 [−0.28, 1.18]	−0.19 [−1.44, 0.59]	−0.78 [−2.61, −0.08]
Broad agreement	58.3%	75.0%	58.3%
Cohen’s κ (broad)	0.000	0.000	−0.250
Exact agreement	33.3%	25.0%	16.7%
Cohen’s κ (exact)	0.053	0.000	−0.111

LLM scores were averaged across 3 iterations within each vendor-application combination before statistical testing. Glass’s Δ uses human SD as the baseline. 95% CIs computed via bias-corrected bootstrap (10,000 resamples). For agreement percentages and Cohen’s κ, each vendor-application combination was reduced to its most frequent recommendation across 3 iterations and this was paired with each individual human review (*n* = 36 pairs for Exp 1; *n* = 24 for Exps 2–3). Bold indicates best alignment.



79.08 $\pm$ 13.32; $p = 0.347$, Glass's $\Delta = 0.34$, 95% CI $(-0.28, 1.18)$], indicating moderate optimism bias. Few-shot prompting (Experiment 2) achieved the best alignment, with LLMs scoring 2.57 points below humans [80.56 $\pm$ 4.81 vs. 83.13 $\pm$ 13.73; $p = 0.671$, Glass's $\Delta = -0.19$, 95% CI $(-1.44, 0.59)$], a non-significant difference representing approximately 3% of the total scale. Adding strict instructions (Experiment 3) produced substantial under-scoring, with LLMs scoring 10.68 points below humans [72.44 $\pm$ 4.98 vs. 83.13 $\pm$ 13.73; $p = 0.097$, Glass's $\Delta = -0.78$, 95% CI $(-2.61, -0.08)$]. Linear mixed-effects models accounting for nested human scores within reviewers did not change any conclusions, they produced similar point estimates but with wider uncertainty [Experiment 1: +4.55, 95% CI $(-9.23, +18.32)$, $p = 0.518$; Experiment 2: -2.57, 95% CI $(-17.04, +11.90)$, $p = 0.728$; Experiment 3: -10.68, 95% CI $(-25.38, +4.02)$, $p = 0.154$]. An exploratory one-way ANOVA restricted to the common application base (FC1 and FC2 only, $n = 6$ per experiment) suggested differences in LLM scores across prompt conditions [$F(2,15) = 8.18$, $p = 0.004$, $\eta^2 = 0.522$]. Given the small group sizes, the exploratory ANOVA result should be interpreted as descriptive rather than confirmatory.

3.2 Criterion-level performance

Criterion-level analyses are exploratory and should be interpreted with caution given the small sample. In the zero-shot condition, the largest deviations were on Methodology (+2.81 points, uncorrected $p = 0.264$) and External Funding Potential (+1.11 points, uncorrected $p = 0.064$); neither reached significance after Holm correction (all corrected $p > 0.38$). Few-shot prompting reduced criterion-level deviations: no criterion showed significant differences in Experiment 2, with the largest

TABLE 3 LLM vendor mean scores and deviation from human mean across experiments.

Vendor	Exp1 deviation	Exp2 deviation	Exp3 deviation
Gemini 2.5 Flash	88.44 (+9.36)	82.33 (-0.79)	78.00 (-5.12)
GPT-5 Nano	77.33 (-1.75)	76.00 (-7.12)	69.83 (-13.29)
Grok 4	85.11 (+6.03)	83.33 (+0.21)	69.50 (-13.62)

deviation being Budget Clarity (-0.99 points). Under strict instructions (Experiment 3), the largest deviation was on Innovation and Impact (-2.61 points, uncorrected $p = 0.031$, Holm-corrected $p = 0.186$). No criterion-level comparison reached statistical significance after correction for multiple comparisons in any experiment.

3.3 Vendor performance

Vendor alignment varied descriptively across conditions (Table 3); no inferential statistics are reported for vendor comparisons given the small sample, and these observations may reflect sampling variability as well as confounding with temperature settings (xAI Grok used temperature = 0.1 vs. 1.0 for other vendors). Because temperature was not standardized across vendors, vendor-level differences in both mean scores and score dispersion cannot be attributed to the models themselves. These results are presented for transparency only. In the few-shot condition, xAI Grok achieved the closest alignment with human scores (+0.21 points from human mean). Google Gemini shifted from the strongest optimism in Experiment 1 (+9.36) to slight under-scoring in Experiment 2 (-0.79). Under

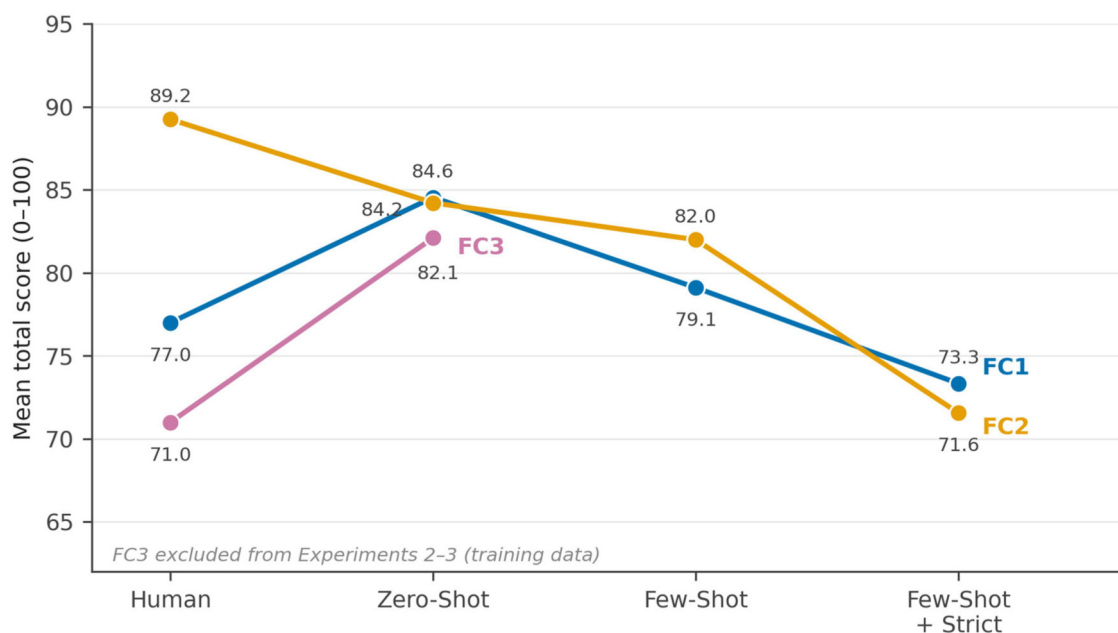


FIGURE 2

Application-level mean total scores by evaluator condition. Lines connect the same application (FC1, FC2, FC3) across the human panel and the three LLM prompt conditions (LLM values are means of vendor-aggregated scores). Humans separated applications by 18.2 points, whereas zero-shot LLM scores compressed all three applications into a 2.4-point range. Human rank ordering (FC2 > FC1) was reproduced only in the few-shot condition. FC3 is absent from Experiments 2–3, in which it served as training data.

strict instructions, all vendors scored below the human mean, with OpenAI (−13.29) and xAI (−13.62) showing the largest deviations.

3.4 Recommendation agreement and consistency

Computed as described in Section 2.5, exact recommendation agreement ranged from 16.7% to 33.3% across experiments, and Cohen's κ -values were at or below zero (exact κ : 0.053, 0.000, −0.111; broad κ : 0.000, 0.000, −0.250 for Experiments 1–3) indicating agreement at or below chance levels. As a sensitivity check, we also computed κ over all raw human-LLM review pairs. These values were similarly near zero (exact κ : 0.061, 0.020, −0.037; broad κ : 0.000, 0.100, −0.071), so the conclusion of chance-level agreement does not depend on how the pairs were formed.

LLMs consistently favored “Fund with Revisions” (52%–78%) compared to humans (25%). In the zero-shot condition, LLMs generated zero “Do Not Fund” recommendations; strict instructions increased this to 27.8% but at the cost of zero “Fund” recommendations. Broad agreement was highest in Experiment 2 (75.0%) compared to 58.3% in Experiments 1 and 3, though κ values indicate this largely reflects chance overlap rather than true agreement.

LLMs demonstrated substantially lower score dispersion across all conditions, with average within-applicant standard deviations of 5.18–5.78 points compared to 11.37–12.56 for humans, representing a 49% reduction in Experiment 1% and 59% in Experiments 2–3 (Figure 1). However, this lower variability likely reflects a compressed scoring range rather than

superior consistency. LLMs did not consistently differentiate between applications: in Experiment 1, LLMs ranked all three applications within a narrow 2.4-point range (82.1–84.6) while humans distinguished a wider 18.2-point spread (71.0–89.2). LLMs matched the human rank ordering of applications only in Experiment 2 (Figure 2).

3.5 Qualitative review feedback

Beyond numerical scores all three LLMs produced criterion-specific written rationales for each review. Table 4 shows representative rationales across models, criteria, and recommendation categories. The rationales named specific strengths and specific gaps that an applicant could act on. GPT-5 Nano, scoring one application on methodology, wrote that “details on the qualitative/data analysis plan and contingency strategies could be more explicit, and the small sample limits generalizability”. Gemini 2.5 Flash flagged sample size directly: “the proposed sample size of 10–12 older adults is very small and will significantly limit the generalizability and statistical power for evaluating the “preliminary impact” (Aim 1), even for a pilot study”. Grok 4 Fast Reasoning tended to pair praise with critique in the same response, as an example: “[t]he mixed-methods pilot design is rigorous, with appropriate pre/post assessments” while also flagging in the same output “[m]inor gaps exist in detailing sample size justification”.

A few patterns held across reviews. First, the models named strengths tied to the specific criterion, and pointed to concrete gaps specific enough to act on. In addition, they appeared to frame their comments within the seed-grant scope rather than comparing the

TABLE 4 Representative examples of LLM-generated criterion-level feedback across models and recommendation categories.

Model	Criterion	Rec.	Representative rationale (Excerpt)
Gemini 2.5 Flash	Methodology (27/30)	Fund	"The mixed-methods pilot design is appropriate, utilizing established and relevant outcome measures for both older adult participants and students. Significant groundwork has already been completed (IRB, space, equipment), and the detailed intervention description and realistic timeline indicate high feasibility for this pilot study".
GPT-5 Nano	Methodology (24/30)	Fund w/ Rev.	"The study uses a mixed-methods pilot with pre/post assessments, a defined 8-week intervention, and multiple validated outcomes, under IRB approval and faculty supervision. [...] Details on the qualitative/data analysis plan and contingency strategies could be more explicit, and the small sample limits generalizability".
Grok 4 Fast	Innovation (25/30)	Fund	"The proposal addresses a significant public health issue—falls among older adults in underserved communities—through an interprofessional student-led program [...]. While the intervention draws on established evidence-based approaches [...], the integration of OT/PT student collaboration in a community-university partnership adds originality".
Gemini 2.5 Flash	Team Strength (5/10)	Do Not Fund	"While the team composition and defined roles (PT, OT, Hospital Admin) are appropriate for the project's aims, the critical omission of biosketches for the PI [...] and two co-investigators [...] severely limits the ability to assess their full qualifications and research track record".
GPT-5 Nano	Budget (8/10)	Fund w/ Rev.	"Budget items are aligned with aims and appear cost-conscious (equipment, education materials, and dissemination). [...] Some line items appear as placeholders (Cost Item X) and the budget presentation has minor formatting inconsistencies, which reduce clarity slightly".

All quotations are verbatim from stored LLM outputs. Bracketed ellipses [...] mark omitted text or edits for de-identification (e.g., removal of investigator names). Examples were selected to span all three models, multiple criteria, and all three recommendation categories.

application to a larger mechanism like an R01. Lastly, even though the numeric scores missed the human rank ordering, the written comments appeared to be potentially useful for future revisions.

4 Discussion

This exploratory pilot study provides case-study-level evidence on the effects of prompt engineering on LLM grant review scoring, based on three applications at a single institution. No experimental condition produced statistically significant score differences from human reviewers after accounting for repeated iterations. Few-shot prompting with training examples drawn from a single application (Experiment 2) produced the smallest deviation from human scores (−2.57 points, $p = 0.671$), while both zero-shot and strict instruction approaches introduced larger biases in opposing directions. On a common application base (FC1 and FC2 only), mean LLM scores ranged from 84.4 (zero-shot) to 72.4 (strict), an 11.9-point spread attributable to prompt condition, underscoring that prompt engineering substantially affects LLM scoring behavior. In the best-aligned experiment (Experiment 2), vendor-level spread (7.3 points) was smaller than the cross-experiment prompt effect (11.9 points), though this pattern was not consistent (Experiment 1 vendor spread: 11.1 points). However, the improved alignment in Experiment 2 cannot be attributed to generalizable few-shot learning. This is due to the training examples coming from a single application, whose human score range (61–80) limits the observed LLM mean (80.56), which makes a simple anchoring explanation to the training scores being equally plausible (see limitations for expanded discussion).

The counterproductive effect of strict instructions (Experiment 3) is noteworthy. Adding directives to “be critical” shifted LLM scores from near-alignment to substantial under-

scoring (10.68 points below human mean), though this difference did not reach statistical significance after correcting for repeated iterations ($p = 0.097$). One speculative explanation is that LLMs may prioritize explicit instructions over the calibrating influence of training examples, a pattern we tentatively describe as instruction compliance bias. We are not aware of any direct evidence for this specific mechanism in the literature, and it cannot be confirmed from our very limited behavioral data alone, so this interpretation should be regarded as speculative. If replicated, this pattern would suggest that providing examples of desired scoring behavior may be more effective than directive language for calibrating LLM outputs.

Arguably the most consequential finding of this study is the dissociation between how well scores aligned and how poorly funding recommendations agreed. Even in the best-aligned condition, LLMs overwhelmingly favored “Fund with Revisions” (77.8% in Experiment 2 vs. 25% for humans), achieved only 25.0% exact agreement, and produced chance-level or below-chance agreement (Cohen’s κ : −0.25–0.05 across experiments). This indicates that observed agreement was largely due to chance overlap rather than true concordance. LLMs may default to intermediate options when uncertain, consistent with documented ambiguity aversion in AI systems (Zhao et al., 2024). The mapping from scores to funding recommendations likely involves implicit thresholds that LLMs cannot infer from rubrics alone.

A potential source of LLM scoring variance is the absence of an explicit operational definition of “seed grant” in our prompts. While human reviewers share institutional understanding of seed grant expectations, LLMs must infer this from training data that likely over-represents larger funding mechanisms (e.g., NIH R01), potentially creating a mismatch in evaluation standards. Future implementations may benefit from including explicit construct definitions alongside rubrics and examples.

4.1 Limitations

Several limitations constrain the generalizability of these findings. Most fundamentally, only three applications from a single institution were evaluated, meaning the study cannot distinguish prompt engineering effects from application-specific effects. The 11.9-point spread in LLM means across prompt conditions (on a common application base) could be confounded with idiosyncratic application characteristics. All statistical tests are underpowered given these effective sample sizes (9 or 6 aggregated LLM observations per experiment), and *p*-values should be interpreted as descriptive guides rather than confirmatory evidence.

The few-shot training examples in Experiments 2 and 3 were derived from reviews of a single application (FC3), with human scores ranging from 61 to 80 points. This makes it difficult to determine whether the LLM learned general scoring behavior, or simply anchored to that particular application's score range. The improved alignment in Experiment 2 could reflect either mechanism. We believe that because of our limited four training scores (61–80) restricting the Experiment 2 LLM mean (80.56), LLM anchoring the range of scores is just as likely as the scenario of LLM “learning” general behavior of a reviewer. Future studies should use training examples drawn from multiple applications to separate these effects.

Additionally, while repeated LLM iterations were addressed through averaging, human scores remain nested within four reviewers who each scored all application which introduced correlation among scores from the same reviewer that the *t*-tests do not account for. The linear mixed-effects models reported in Section 3.1 address this nesting, and left all conclusions unchanged. Cohen's κ produced similarly near-zero values, so the conclusion of chance-level agreement is robust to the data structure.

The observed 49%–59% reduction in LLM score variability should be interpreted cautiously. LLMs compressed scores into a narrow range and failed to rank applications in the same order as humans in two of three experiments, suggesting that this lower variability reflects a narrower scoring range and reduced sensitivity to differences in application quality rather than superior consistency. The low-temperature setting for xAI Grok (0.1) may further contribute to reduced variability for that vendor. The seed grant context (\$5,000) may not generalize to larger, more complex grants. Model versions tested (October 2025) may perform differently from current versions. The retrospective design and lack of preregistration should also be noted. Despite these limitations, the systematic experimental design with appropriate controls for data contamination provides a methodological framework for future, larger-scale studies.

4.2 Implications and future directions

These findings generate several hypotheses for future testing. First, the observation that few-shot prompting without directive language produced the best-aligned scores

warrants replication with larger application pools and training examples drawn from multiple applications. Second, the gap between score calibration and recommendation agreement suggests that these may require separate calibration strategies in future implementations. Third, the near-zero kappa values across all conditions indicate that LLM-generated funding recommendations do not yet provide meaningful signal beyond chance, reinforcing that any future hybrid model must retain human authority over funding decisions. Future research should employ larger, multi-institutional application samples, mixed-effects models to properly account for the nested data structure, and preregistered designs to enable confirmatory inference.

One final contribution of this work sits outside score alignment: the written rationales themselves. Internal grant review, especially for seed grants, often returns only a score with no written feedback. In this study, all three LLMs produced structured, criterion-specific comments that named strengths and specific weaknesses (Table 4). The comments pointed to concrete elements of the applications like sample size, budget allocation, and missing material. We cannot say from these data whether applicants would find the feedback useful, or whether acting on it would improve the next submission. What we can say is that generating this kind of feedback costs essentially nothing, and for earlier-career investigators still developing their grantsmanship, that matters. A useful next step is to ask applicants directly how they judge the quality and usefulness of LLM-generated feedback.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

HW: Funding acquisition, Validation, Conceptualization, Resources, Writing – review & editing, Visualization, Project administration, Supervision, Writing – original draft, Methodology, Formal analysis, Software, Data curation, Investigation. JL: Writing – original draft, Resources, Validation, Conceptualization, Methodology, Writing – review & editing, Data curation. EL: Writing – original draft, Funding acquisition, Resources, Conceptualization, Validation, Visualization, Writing – review & editing, Project administration, Investigation, Supervision, Formal analysis, Methodology, Data curation, Software.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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The author(s) declared that generative AI was used in the creation of this manuscript. Generative AI (Anthropic Claude) was used to assist with editing and refining portions of the manuscript text. All AI-assisted content was reviewed, verified, and revised by the authors to ensure accuracy and scholarly integrity. The authors take full responsibility for the final content.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/feduc.2026.1856134/abstract#supplementary-material>

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